Lesson 13 More on Combinatorial Proofs

For each of the following identities, give a combinatorial proof.

1. Let $n \in \mathbb{N}$, then
$$\binom{2n}{2} = 2\binom{n}{2} + n^2$$
2. Let $n \in \mathbb{N}$, then
$$n^2 = n(n-1) + n$$
3. Let $n, k \in \mathbb{N}$ and $0 \le k \le n$, then
$$P(n,k) = P(n-1,k) + kP(n-1,k-1)$$
4. Let n and k be integers and $2 \le k \le n$, then
$$k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$$
Hint: Consider choosing a team with a captain and a vice-captain.
5. Let n, k, l be nonnegative integers and $0 \le k + l \le n$, then
$$\binom{n}{k}\binom{n-k}{l} = \binom{n}{k+l}\binom{k+l}{l}$$
6. Let $n \ge k$, then
$$n! = \binom{n}{k} k!\,(n-k)!$$
Hint: To arrange n objects, we may arrange k of them first.
7. Let $n \ge 3$, then
$$\binom{n}{3} = \binom{2}{2} + \binom{3}{2} + \cdots + \binom{n-1}{2}$$
Hint: Mimic the proof of Proposition 5 in Lesson 13.
8. Let $n \in \mathbb{N}$, then
$$\binom{2n+2}{n+1} = \binom{2n}{n+1} + 2\binom{2n}{n} + \binom{2n}{n-1}$$
Hint: Consider a set with two elements a and b, and $2n$ other elements. How many ways are there to construct a subset of $n+1$ elements?
9. Let n, r, k be nonnegative integers and $0 \le k \le r \le n$, then
$$\binom{n}{r}\binom{r}{k} = \binom{n}{k}\binom{n-k}{r-k}$$